

MathD2: Towards Disambiguation of Mathematical Terms

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Fifth Workshop on Scholarly Document Processing
(@ ACL 2025, Vienna, Austria, July 31, 2025)

Building a Dictionary of Mathematical Terms in Papers

Same *path* and different *blocks* extracted from scholarly papers

Definiendum	Definition in Source Article
block	A <i>block</i> in H is a maximal set of tightly-connected hyperedges. (Ergemlidze et al., 2019)
block	A <i>block</i> of indices is a set of numbers S where every term $SG_{a,b}(s)$ depends on the same value via division, for all $s \in S$. (Kupin, 2011)
path	If the vertices $v_0, v_1, \dots, v_k$ of a walk W are distinct then W is called a <i>Path</i> . A path with n vertices will be denoted by P_n . P_n has length $n - 1$. (Kalayathankal et al., 2015)
path	Let $G = (V, E)$ be a graph. A <i>path</i> in a graph is a sequence of vertices such that from each of its vertices there is an edge to the next vertex in the sequence. This is denoted by $P = (u = v_0, v_1 \dots, v_k = v)$, where $(v_i, v_{i+1}) \in E$ for $0 \leq i \leq k - 1$. (Perera and Mizoguchi, 2012)

Definitions extracted from different scholarly articles (Jiang and Senellart, 2023).

ProofWiki Disambiguation Pages

Definition:Path

Disambiguation

This page lists articles associated with the same title. If an internal link led you here, you may wish to change the link to point directly to the intended article.

Path may refer to:

- **Topology:**
 - **Path:** a **continuous mapping** $f : [a . . b] \rightarrow T$.
- **Complex Analysis:**
 - **Path (Complex Plane).**
- **Graph Theory:**
 - **Path:** a **trail** in which all **vertices** are distinct.
 - **Path** in a **Digraph:** as a **path** in an **undirected graph** but taking into account the direction of the **arcs**

- Manually built
 - Provides few equivalent mathematical definitions
 - Mixture of formulas and natural language
 - Limited number of disambiguation pages
- **RQ1:** How well can contextualized text representation help the disambiguation of mathematical terms?
- **RQ2:** Which architectures and pretraining strategies are best suited?
- **RQ3:** How well can LLMs do it?

MathD2 Dataset

ProofWiki Disambiguation Pages

→ 365 ambiguous terms, 1,984 definition-title pairs

Definition:Path (Topology)

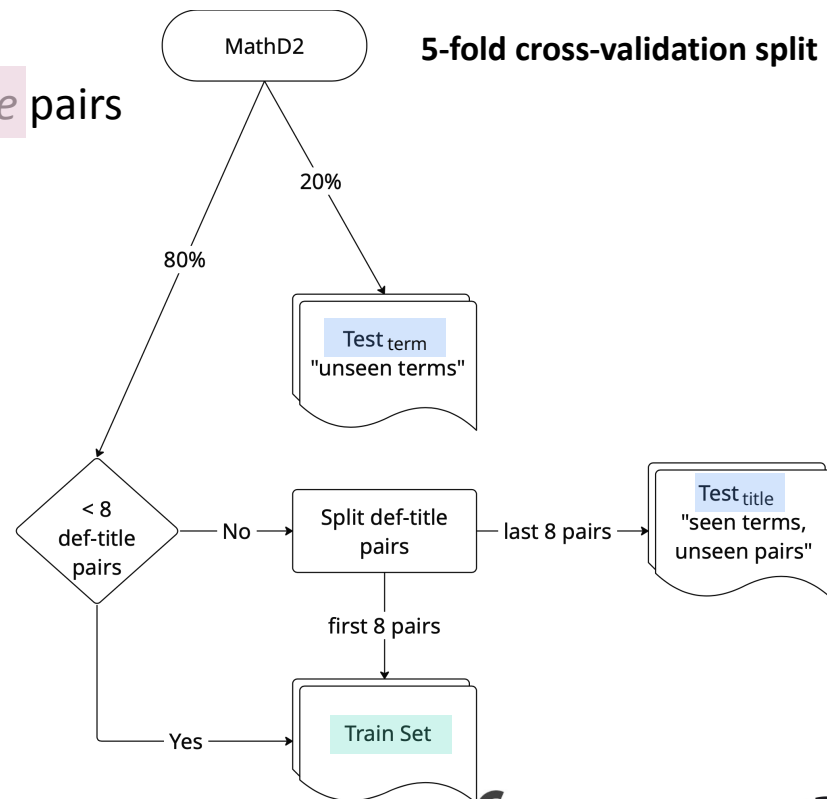
This page is about **path** in the context of **topology**. For other uses, see **path**.

Definition

Let $T = (S, \tau)$ be a topological space.

Let $I \subset \mathbb{R}$ be the closed real interval $[a . . b]$.

A **path** in T is a continuous mapping $\gamma : I \rightarrow S$.



Entity Linking for Definiendum Disambiguation

General idea:

- Input: Definition + Candidate Title (definiendum, domain)
- The proposed approach produces a score for each candidate input.
- The title with the highest score is selected as the final predicted output.

Definition	Candidate Title	Score
A bilinear form is a linear form of order 2.	Definition: Bilinear Form (Linear Algebra)	0.6
	Definition: Bilinear Form (Polynomial Theory)	0.8



predicted

Evaluation: predicted title == true title?

Approach 1: Using Next Sentence Prediction (NSP)

Input: (Title, Definition) **Output:** NSP(BERT ([CLS] Title [SEP] definition [SEP]))

Finetuning data:

Positive samples:

[CLS] Title A [SEP] Definition A [SEP]

[CLS] Title B [SEP] Definition B [SEP]

Negative samples:

[CLS] Title A [SEP] Definition B [SEP]

[CLS] Title B [SEP] Definition A [SEP]

Objective: Bring the positive pairs together, and to push apart the negative pairs

Training loss:

$$L = \max \{0, \lambda - d_{\text{neg}} + d_{\text{pos}} \}$$

Approach 2: Using Semantic Text Similarity (STS)

Input: (Title, Definition) **Output:** $\text{cosine}(\text{SBERT}(\text{Title}), \text{SBERT}(\text{Definition}))$

- The embeddings of titles and definitions only need to be calculated once.
- Cosine similarity calculation is faster than NSP inference.
- Limited pretrained models for mathematical text comparison.

Approach 3: Instruction-Tuned LLMs for Term Disambiguation

1. Task Description

"Your task is to find the correct article title given a *mathematical term* and *definition* as context."

2. Hallucination Prevention

"Reply with "I don't know" when uncertain."

3. Expectation Setting

"Only select one answer from the provided list. Do not provide justifications."

4. Multiple Choice

"Identify the correct *title* from this list:[...]"

Experimental Setup → Zero-Shot

Results & Discussion

Model	Approach	Test _{term}	Test _{title}
BERT (Devlin et al., 2019)	NSP	84.6	84.0
BERT finetuned with MathD2	NSP	92.3	91.6
BERT mean pooled	STS	39.9	27.2
BERT pretrained with papers (Mishra et al., 2021)	STS	44.0	32.7
SBERT-all-mpnet-base-v2 (Reimers and Gurevych, 2019)	STS	92.8	91.9
SBERT-all-MiniLM-L6-v2	STS	91.6	91.4
SBERT-all-MiniLM-L12-v2	STS	92.6	91.4
SBERT pretrained with paper titles and abstracts (Steinfeldt and Mihaljević, 2024)	STS	59.8	48.4
Mistral-7B-Instruct-v0.2 (Jiang et al., 2023)	LLM-Instruct	45.9	50.4
Mistral-7B-Instruct-v0.3	LLM-Instruct	60.0	52.1
Meta-Llama-3-8B-Instruct (AI@Meta, 2024)	LLM-Instruct	75.0	71.9



MathD2 is useful for finetuning !

Table 4: Averaged accuracy scores of five tests. Values are reported as $p \cdot 100$. Best scores are in bold.

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★ Domain adaptation with Masked Language Modeling (MLM) slightly improves STS.

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- ★ But SBERT's pretraining is more efficient!
- ★ Adapted SBERT with title-abstract pairs cannot generalize to our task.

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★ LLMs tend to not follow instructions.

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★ The difference between finetuned NSP and STS is not statistically significant.

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Limitations

Common errors of both NSP & STS, e.g. when there are common/related words in the candidate title and the definition

Definition

Let G be a **game**.

Let P be a **player** of G .

Let A be the **set** of **moves** available to P .

Let C be the **set** of **consequences** of those **moves**.

A **consequence function** for P is a **mapping** from the **set** A to the **set** C :

$$g : A \rightarrow C$$

https://proofwiki.org/wiki/Definition:Consequence_Function

Candidate titles:

Definition: Consequence Function

Definition: Consequence
(**Game** Theory)

predicted

Limitations - LLM Errors

Inspected errors from ***test_{term}***:

1. No Prediction (NP).

- LLM responds with “I don’t know.”

2. Not Following Instructions (NFI).

- Non-multiple choice answers.

3. Learning Bias (LB).

- Answers in prose, and non-categorical.

4. Wrong Prediction (WP).

- Incorrect choice.

Error Type	Mistralv2	Mistralv3	Llama3
NP	9.72 %	0.00 %	2.27 %
NFI	78.97 %	78.09 %	73.17 %
LB	1.55 %	0.73 %	0.60 %
WP	9.76 %	21.18 %	23.96 %

Future Work

- Finetune SBERT model for mathematical text comparison with different tasks.
- Combine textual representation and bibliographic information.
- Create a larger entity linking dataset with definitions in scholarly papers.

Thank you for your attention!

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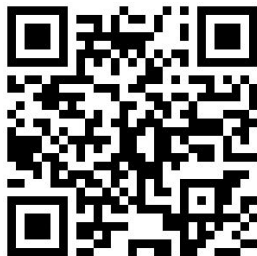
Slides:

<https://t.ly/1p-77>



Code & Data:

<https://t.ly/Dgqdz>



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Karlsruher Institut für Technologie

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